

Arbitrary-dimensional CPWL density in Hessian–Schatten energy

Literature answer to Conjecture 1 in arXiv:2210.04077

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Status: literature-already-answered, full intended conjecture. For every d , continuous piecewise-linear functions are dense, in $L^1((0,1)^d)$ and in Schatten-1 Hessian energy, among functions of bounded Hessian–Schatten variation. Ambrosio–Brena–Conti explicitly identify their Theorem 2.4 as a positive answer to the source’s Conjecture 1.

Source conjecture

Ambrosio–Aziznejad–Brena–Unser prove in Theorem 21 that, for $\Omega = (0,1)^2$, every function f of bounded Hessian–Schatten variation has CPWL approximants f_k satisfying

$$f_k \longrightarrow f \quad \text{in } L^\infty(\Omega), \quad |D_1^2 f_k|(\Omega) \longrightarrow |D_1^2 f|(\Omega).$$

Their Conjecture 1, on source PDF page 14, asks whether the density result extends to every hypercube $(0,1)^n$ [1]. The arXiv v1 wording says only that Theorem 21 should remain valid in arbitrary dimension. The published version makes the intended formulation precise: in dimensions $n \geq 3$, the L^∞ topology is replaced by L^1 (and it notes $L^{n/(n-2)}$ as another natural ambient topology). Its conjecture footnote also records Sergio Conti’s announced proof.

Explicit later solution

Ambrosio–Brena–Conti state immediately before Theorem 2.4, on supporting PDF page 14, that the theorem “gives a positive answer” to the source’s Conjecture 1 [2]. Their precise result is:

For every d and $\Omega = (0,1)^d$, if $f \in L^1_{\text{loc}}(\Omega)$ has bounded Hessian–Schatten variation, then there are $f_k \in \text{CPWL}(\Omega)$ such that

$$f_k \longrightarrow f \quad \text{in } L^1(\Omega), \quad |D_1^2 f_k|(\Omega) \longrightarrow |D_1^2 f|(\Omega).$$

This is exactly the published arbitrary-dimensional conjecture. The supporting authors knew the identification: they cite Conjecture 1 and Theorem 21 of the source explicitly. This is therefore a retrieved literature answer, not an agent-generated theorem.

Mechanism and scope

The supporting paper first proves its global Theorem 2.2 for smooth targets in arbitrary dimension. It constructs triangulations whose local orientations follow the eigendirections of the target Hessian and controls the energy cost of gluing differently oriented local meshes. An extension lemma, mollification, and a diagonal argument then yield the localized Theorem 2.4 for arbitrary finite-energy targets.

The answer is full for the intended L^1 formulation. It should not be read as an L^∞ density theorem in dimensions $d \geq 3$. In dimension two, the source’s L^∞ result remains stronger in its function topology.

Evidence checked

The identification was checked against the local arXiv sources and rebuilt PDFs for both papers, as well as the publishers' official article text. The decisive locations are Conjecture 1 on source PDF page 14 and Theorem 2.4, including its explicit positive-answer sentence, on supporting PDF page 14.

References

- [1] L. Ambrosio, S. Aziznejad, C. Brena, and M. Unser, *Linear Inverse Problems with Hessian–Schatten Total Variation*, arXiv:2210.04077; Calc. Var. Partial Differential Equations **63**, article 9 (2024), doi:10.1007/s00526-023-02611-6.
- [2] L. Ambrosio, C. Brena, and S. Conti, *Functions with Bounded Hessian–Schatten Variation: Density, Variational, and Extremality Properties*, arXiv:2302.12554; Arch. Ration. Mech. Anal. **247**, article 111 (2023), doi:10.1007/s00205-023-01938-w.